

May 15, 2020



U.S. Department
of Transportation

East Building, PHH-30
1200 New Jersey Avenue S.E.
Washington, D.C. 20590

**Pipeline and Hazardous
Materials Safety Administration**

DOT-SP 20994

EXPIRATION DATE: 2022-04-30

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: SK Innovation Co. Ltd
Seoul, South Korea

US AGENT: SK Innovation Detroit Office
Troy, MI

2. PURPOSE AND LIMITATION:

a. This special permit authorizes the transportation in commerce of lithium batteries exceeding 35 kg by cargo-only aircraft. This special permit provides no relief from the Hazardous Materials Regulations (HMR) or the International Civil Aviation Organization's Technical Instructions for the Safe Transport of Dangerous Goods by Air (ICAO TI) other than as specifically stated herein. The most recent revision supersedes all previous revisions.

b. The safety analyses performed in the development of this special permit only considered the hazards and risks associated with the transportation in commerce.

c. This special permit serves as an approval under Special Provision A99 of the ICAO TI as a "Competent Authority Approval" as defined under 49 CFR §107.1.

d. No party status will be granted to this special permit.

3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.

Tracking Number: 2020014657

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4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR 172.101 Column (9B) in that lithium ion batteries may not exceed 35 kg when transported by cargo aircraft, except as provided herein.
5. BASIS: This special permit is based on the application of SK Innovation Co. Ltd dated January 13, 2020 and supplemental information dated February 27, 2020, submitted in accordance with § 107.105 and the public proceeding thereon.
6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Lithium ion batteries <i>including lithium ion polymer batteries</i>	9	UN3480	N/A

7. SAFETY CONTROL MEASURES:

a. OPERATIONAL CONTROLS:

(1) This special permit is valid only for transportation of the lithium ion battery modules (comprised of SK P702 cells) identified as:

(i) 5P11S (nominal energy rating of 16.1 kWh);

(ii) 4P10S (nominal energy rating of 11.7 kWh);
and

(iii) 5P8S (nominal energy rating of 11.7 kWh).

(2) Cells and battery modules must be protected against short-circuiting.

(3) Battery modules must be equipped with an effective means to prevent dangerous reverse current flow for cells connected in parallel.

(4) All of the battery modules must be offered for transportation at a state of charge not greater than 30%.

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b. TESTING REQUIREMENTS: Each cell and battery module must be a type that has passed the required tests in accordance with the UN Manual of Tests and Criteria, 6th Revised Edition.

c. PACKAGING:

(1) Inner Packaging: Each battery must be placed in a non-metallic inner packaging that completely encloses the battery, and separates the battery from contact with conductive materials in the packaging.

(2) Outer Packaging: The battery module must be packaged in an outer packaging consisting of a 4C1 wooden box meeting the Packing Group I performance level.

(3) Packaging must comply with the requirements in ICAO TI Packing Instruction 965.

(4) The nominal net weight of each 5P11S, 4P10S, and 5P8S battery module is 68 kg, 50 kg, and 50 kg, respectively and may not exceed these limits within a 3 % variation.

(5) Only one (1) battery module may be packaged in a package.

d. MARKING REQUIREMENTS: Each package, and overpack if used, prepared under the provisions of this approval must be plainly marked with the approval number "DOT-SP 20994".

8. SPECIAL PROVISIONS:

a. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modification or change is made to the package or its contents and it is reoffered for transportation in conformance with this special permit and the HMR.

b. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

c. A current copy of this special permit must be maintained and made available for examination at each

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facility where the materials are packaged and offered or reoffered for transportation.

d. This special permit in no way affects the need to obtain any required authorizations from other agencies of the United States Government or from the competent authorities of countries of origin, transit, and destination.

e. The special permit holder must maintain a record of all activity conducted under the authority granted in this special permit.

f. The record must contain a complete listing and number of shipments made to include:

(1) Dates of Shipment.

(2) Description of each type of shipment (to include origination/destination).

g. All of the above information must be made available upon request to a DOT representative or an enforcement Official.

h. A battery or monitoring system that is determined to be defective may not be shipped under this special permit.

9. MODES OF TRANSPORTATION AUTHORIZED: Cargo-aircraft only.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.

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- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) – "The Hazardous Materials Safety and Security Reauthorization Act of 2005" (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term "exemption" to "special permit" and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

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Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-30, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: Steve H/Andrew Eckenrode